

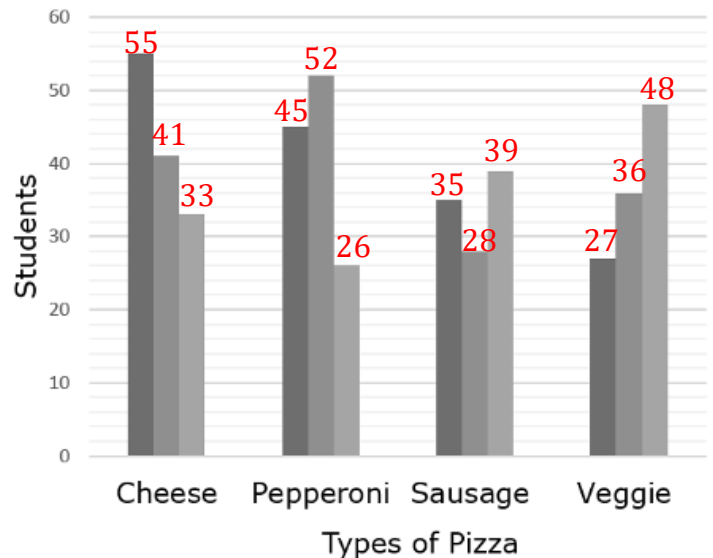
College students were polled about their favorite type of pizza. They were given a choice of cheese, pepperoni, sausage, and veggie. The results are shown in the table below. Use this information to complete problems 1-5. Round to the nearest whole percent.

1. What is the probability that a randomly chosen junior will pick cheese as their favorite type of pizza?

$$\frac{33}{146} \approx 0.23 \approx 23\%$$

2. What is the probability that a randomly selected student that is a freshman or Sophomore will pick cheese as their favorite type of pizza?

$$\frac{55+41}{162+157} = \frac{96}{319} \approx 0.30 \approx 30\%$$



162 ■ Freshman  
157 ■ Sophomores  
146 ■ Juniors

3. The probability of a sophomore choosing sausage pizza is about 15 %

☐ more  
☒ less

than a sophomore choosing pepperoni pizza.

Sausage:  $\frac{28}{157} \approx 18\%$       Pepperoni:  $\frac{52}{157} \approx 33\%$        $33 - 18 = 15$

4. The probability of a sophomore choosing veggie pizza is about 10 %

☐ more  
☒ less

than a junior choosing veggie pizza.

Sophomore:  $\frac{36}{157} \approx 23\%$       Junior:  $\frac{48}{146} \approx 33\%$        $33 - 23 = 10$

5. A pizza vendor decides to bring more cheese than pepperoni pizza to sell at freshman orientation. Explain whether this a good business decision based on the graph.

Cheese  
Cheese  $\frac{129}{465} \approx 28\%$   
total

Pepperoni  
 $\frac{123}{465} \approx 26\%$

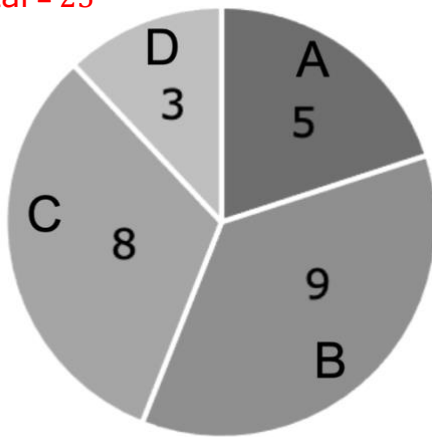
This is a good decision because slightly more students favor cheese over pepperoni.

6. Match the probabilities with the event described. Draw a line from the event to the probability of it occurring.

| Event                                                                                                                                                                          | Probability that the Event will Occur |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| A six-sided dice lands on a 4. $\frac{1}{6} \approx 16.7\%$                                                                                                                    | 50%                                   |
| A class had 16 girls and 18 boys. What is the probability that the teacher picks a girl to lead the class to lunch? $\frac{16}{34} \approx 47\%$                               | 16.7%                                 |
| A bag has 10 purple marbles, 6 black marbles, and 4 red marbles. If marbles are chosen at random, what is the probability that a purple marble is chosen? $\frac{10}{20} = .5$ | 47.1%                                 |

Students in Mrs. Dobson's class took their Unit 3 Science test on Monday. The pie graph shows the number of students that earned each grade. Use the information in the graph to answer questions 7-10.

Total = 25



7. What percentage of students earned a D?

$$\frac{3}{25} = 0.12 = 12\%$$

8. What percentage of students earned a B?

$$\frac{9}{25} = 0.36 = 36\%$$

9. If Mrs. Dobson's students score in the same manner for the Unit 4 test, what is the probability that students will not earn an A or B?

$$\frac{11}{25} = 0.44 = 44\%$$

10. Based on the data, explain whether students should be worried about passing the Chapter 3 test with a grade of C or higher?

$$\frac{8 + 9 + 5}{25} = \frac{22}{25} = 0.88 = 88\%$$

No, they should not be concerned because 88% of students scored a C or higher.